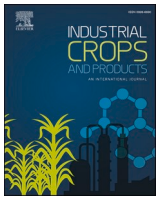
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Structure and functionality of cassava starch in different deep eutectic solvents/water mixtures: A comparative study

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ABSTRACT

Starch, as one of the most abundant natural polysaccharides, is regarded as a sustainable material due to its renewability, biodegradability, and modifiability, and is widely used as an excellent industrial material. However, as a polyhydroxyl polymer, native starch has low solubility in most situations due to its network structure formed by intramolecular and intermolecular hydrogen bonds, which limits its industrial application. In this study, the solubility of cassava starch (CS) in a series of acidic, neutral, and alkaline deep eutectic solvents (DESs) was investigated. Relatively high solubilities of 52.8 wt%, 11.4 wt%, and 10.2 wt% were obtained in choline chloride/formic acid (CF), ethylammonium chloride/ethylene glycol (EE), and choline chloride/diethanolamine (CD), respectively. The dissolution behavior of CS in pure mixtures of these DESs and in DESs/water mixtures (2:8) was also compared. The results of gel permeation chromatography (GPC) analysis showed that the weight-average molecular weights (Mw) and number-average molecular weights (Mn) were ranked in the order CF > EE > CD. The same pattern was observed for the DESs/water mixtures (2:8), although the reduction in Mw and Mn was less distinct than in the pure DESs. The results of X-ray diffraction, small-angle X-ray scattering, and scanning electron microscope observations were consistent with the GPC data, i.e., the hydrolysis ability of CS was higher in the acidic DESs (CF) compared to the neutral (EE) and alkaline (CD) DESs. The dissolution behaviors of CS in the DESs/water mixtures were affected by the structure of the DESs and their water content. These results demonstrate that aqueous DESs serve as excellent media for starch dissolution. Moreover, this study provides a comprehensive understanding of the solubilization behavior of CS in different aqueous DESs and provides insights into the development of novel starch-based materials using DESs.

1. Introduction

Second only to cellulose, starch is the most renewable raw material in nature; it is a natural polymer that is widely available, biocompatible, and inexpensive (Santander-Ortega et al., 2010). Native starch is a semi-crystalline polymer mainly composed of two types of α-glucan, linear amylose, and highly branched amylopectin (Fan, 2015; Lan et al., 2016; Li and Wei, 2020). It has multiple hydroxyl groups and forms intra- and intermolecular hydrogen bonds (Huber and Bemiller, 2000), thus creating complex hydrogen-bonded network structures, and regular and orderly crystalline structures among polysaccharide chains of starch. These structures lead to low starch solubility in many conventional organic solvents (Radosta et al., 2001), which limits starch modification and its industrial application. Indeed, although starch dissolves in several solvents, such as dimethyl sulfoxide (DMSO) (Jackson, 1991; Schmitz et al., 2009), dimethylacetamide/LiCl (Heinze et al., 2000; Huang and Yu, 2010; Koganti et al., 2011), ionic liquids (Koganti et al., 2011), KOH, and NaOH (Han and Lim, 2004), high temperatures or other special conditions are typically required.

Novel and "green" solvents, such as deep eutectic solvents (DESs)

Abbreviations: ATR-FTIR, attenuated total reflection-Fourier transform infra-red; CD, choline chloride/diethanolamine; CDW, CD/water; CF, choline chloride/formic acid; CFW, CF/water; CS, cassava starch; DESs, deep eutectic solvents; DMSO, dimethyl sulfoxide; EE, ethylammonium chloride/ethylene glycol; EEW, EE/water; GPC, gel permeation chromatography; Mn, number-average molecular weight; Mw, weight-average molecular weight; SAXS, small-angle X-ray scattering; SEM, scanning electron microscope; XRD, X-ray diffraction.

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Table 1
Preparation of pure DESs and determination of their viscosity and starch solubility.

DES[a]	HBA[b]	HBD[c]	Abb.	Mol/mol	Viscosity[d] (mPa·s)	Solubility[e] (wt%)
Acid	ChCl	Lactic acid	CL	1:2	53.13	11.6%
				1:5	53.41	12.2%
				1:10	53.34	11.9%
	ChCl	Oxalic acid	CO	1:1	108.09	10.6%
				1:2	101.29	10.9%
	ChCl	Formic acid	CF	1:2	11.31	41.5%
				1:3	11.07	41.7%
				1:4	10.85	52.8%
Neutral	ChCl	Acetic acid	CA	1:2	20.07	11.0%
	ChCl	Glycerine	CG	1:3	23.14	12.1%
	ChCl	Ethylene glycol	CE	1:4	25.06	12.3%
				1:2	114.48	6.1%
				1:3	116.83	6.0%
				1:2	60.1	0%
	EhCl	Glycerine	EG	1:2	51.5	0%
	EhCl	Ethylene glycol	EE	1:2	38.6	11.4%
				1:3	42.1	0%
				1:4	46.7	0%
Alkaline	ChCl	Diethanolamine	CD	1:6	93.9	0%
				1:8	106.7	10.2%
				1:10	112.6	2.17%
	ChCl	Urea	CU	1:2	132.5	0%
	ChCl	Acetamide	CAT[f]	1:2	/[g]	0%

[a] DES, deep eutectic solvents.
[b] HBA, hydrogen bond acceptor.
[c] HBD, hydrogen bond donor.
[d] Viscosity was measured at 22–23 °C.
[e] Starch solubility was measured after reaction at 100 °C for 2 h.
[f] DES status is solid.
[g] "/" indicates that the viscosity could not be measured.

(Cao et al., 2019; Ren et al., 2019a; Zdanowicz, 2020; Zhang et al., 2017), have been considered for their potential ability to dissolve starch as, as such, their potential to support the development of starch-based materials (Ren et al., 2020). DESs are a new class of ionic liquid (IL) analogs consisting of hydrogen bond acceptors and hydrogen bond donors (Abbott et al., 2007). These not only possess similar reaction properties as ILs but have the advantages of low cost, high atomic utilization, simple preparation requirements, and non-volatility (Smith et al., 2014). As such, DESs have been identified as potential alternative green solvents for polysaccharides including starches (Ren et al., 2020, 2019b; Zhang et al., 2017), cellulose (Dadi et al., 2010), lignin (Soares et al., 2017), and even biomass (Chen and Mu, 2019). Biswas et al. (2006) showed that the most complete starch dissolution was achieved in a mixture of choline chloride and urea (solubility: 9.1 wt%), and choline chloride and citric acid (solubility: 8.3 wt%) after heating at 100 °C, and the use of 1-butyl-3-methylimidazolium chloride achieved solubility of up to 15 wt% at 80 °C. Similar results were obtained for potato starch added to choline chloride/urea, choline chloride/citric acid, and choline chloride/succinic acid solutions for 60 min at temperatures ranging 118–135 °C (Zdanowicz and Spychaj, 2011). More recently, the effects of water on ILs, such as polarity, viscosity, and solvating ability, have been examined (Kohno and Ohno, 2012; Zhao et al., 2015). It has also been shown that the presence of water promotes the formation of hydrogen bonds between Ac^- and starch hydroxyls; a 0.15:1 mol/mol [C_2mim][OAc]:water mixture effectively dissolved starch under ambient conditions without derivatization or degradation of the starch (Zhang et al., 2015, 2017). Furthermore, Lu et al. (2020) found that the IL/H_2O ratio strongly influenced the appearance and microstructure of maize starch and potato starch, and water/IL mixtures offer better dissolution ability for starch than pure ILs.

Despite this existing work, few studies have considered a comprehensive comparative analysis of starch dissolution in acidic, neutral, and alkaline aqueous DESs. Inspired by this, in this study, after screening 11 kinds of DESs composed of different hydrogen bond receptors and hydrogen bond donors for starch dissolution, we selected the following three DESs with the highest starch dissolution rates and investigated the dissolution behavior of cassava starch (CS): (1) choline chloride/formic acid (CF), (2) ethylammonium chloride/ethylene glycol (EE), and (3) choline chloride/diethanolamine (CD). The effect of different DESs/water ratios on the dissolution behavior of CS was also examined based on an analysis of structural changes in the starch after dissolution. Specifically, the properties and mechanisms of dissolution were studied using polarized light microscopy, scanning electron microscope (SEM) observations, X-ray diffraction (XRD), small-angle X-ray scattering (SAXS), attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectroscopy, and gel permeation chromatography (GPC). In doing so, we provide a more comprehensive understanding of the dissolution behavior of CS in different aqueous DESs than has been previously available.

2. Materials and methods

2.1. Materials

Cassava starch (0.2% moisture) was obtained from the Hongfeng Starch Co., Ltd. (Guangxi, China); choline chloride (AR, 98%), ethylamine hydrochloride (AR, 98%), and ethylene glycol (AR, 98%) were purchased from the Aladdin Industrial Co., Ltd., China; formic acid (AR, 88%) was purchased from the FUYU Fine Chemical Co., Ltd., Tianjin, China; and diethanolamine (AR, 99%) was purchased from the Damao Chemical Reagent Factory (Tianjin, China). Deionized water was used in all the experiments.

2.2. Preparation and screening of DESs

The components of the DESs were vacuum-dried at 80 °C for 5 h prior to use and then mixed in glass vials according to the corresponding molar ratios. They were then sealed and placed in an oil bath (60 °C) with magnetic stirring until a homogeneous clear liquid was obtained. A detailed description of the prepared DESs is provided in Table 1. Native

CS was added to the prepared DESs in 0.05 g increments at 100 °C for 2 h in an oil bath. Maximum starch solubility was determined when the DESs-treated samples showed a dark field under polarized light microscopy and no crossover point was observed.

Three DESs (acidic, neutral, and alkaline) were selected (Table 1) for the CS dissolution experiment according to the maximum starch solubility, namely ChCl:formic acid (CF), EhCl:ethylene glycol (EE), and ChCl:diethanolamine (CD) at molar ratios of 1:4, 1:2, and 1:8, respectively.

2.3. Measurement of solvatochromic parameters of DES/water mixtures

N, N-diethyl-4-nitroaniline (DENA), 4-nitroaniline (NA), and Nile Red (NR) dyes were used to measure the solvatochromic parameters of the DESs/water mixtures (Oh et al., 2020). For this, a 0.1 mM dye stock solution was prepared in methanol, and 5 mL of the dye stock solution was transferred to a centrifuge tube and heated at 45 °C for 24 h to remove the methanol. After adding 3 mL of the DES/water mixtures to the centrifuge tube, sonication was performed at 25 ℃ to dissolve the dye. The absorption spectrum of the mixture was then determined by UV-Vis spectroscopy at 25 ℃ in the 300–700 nm wavelength range using a SpeCond plus 50 (Analytikjena, Germany). The soluble discoloration parameters were calculated using the following equations (Xue et al., 2016):

$$E_{NR} = 28591/\lambda_{max} \quad (1)$$

$$\pi^* = 0.314 \times (27.52 - \nu_{DENA}) \quad (2)$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \nu_{NR}}{1.6078} \quad (3)$$

$$\beta = 0.358 \times (31.10 - \nu_{NA}) - 1.125 \times \pi^* \quad (4)$$

where λ_{max} is the wavelength corresponding to the maximum absorption of Nile Red, and $\nu_{NR} = 1/(\lambda_{(NR)max} \times 10^{-4})$, $\nu_{DENA} = 1/(\lambda_{(DENA)max} \times 10^{-4})$, and $\nu_{NA} = 1/(\lambda_{(NA)max} \times 10^{-4})$.

2.4. Starch treatment in DES/water mixtures

Mixtures of the three DESs and water with different mass ratios (2:8, 4:6, 5:5, 6:4, 8:2, and 10:0) were prepared, referred to as CFW, EEW, and CDW, respectively, and where DESs/water-10:0 refers to pure DESs with no water added. Native CS (5 wt%) was added to the DESs/water mixtures and then placed in a capped 50-mL conical flask in an oil bath for 2 h at 25 °C, 60 °C, 80 °C, and 100 °C with magnetic stirring. Subsequently, ten volumes of ethanol were added with stirring. The mixtures were centrifuged at 4500 rpm for 15 min and the precipitated starch was washed three times with ethanol to remove the residual DESs. Then, for each sample, the liquid was decanted and the separated solid placed in a vacuum desiccator at 45 °C for 24 h. The solids were ground and passed through a 200-mesh sieve to obtain the regenerated CS. The regenerated CS yield was calculated using the following equation:

$$\text{Regenerated CS yield}(\%) = \frac{Mass\ of\ the\ regenerated\ CS}{Mass\ of\ the\ native\ CS\ in\ the\ initial\ sample} \times 100 \quad (5)$$

2.5. Recyclability of DESs/water mixtures

Recycling experiments were prepared using a distillation condensing unit following a previously reported procedure (Yuan et al., 2017). The mixtures of starch/DESs/water were centrifuged at 4500 rpm for 15 min, and the supernatants were distilled and condensed at 100 °C to remove the water. The liquor collected after concentration, referred to as the 'recovered DESs', was used for the next cycle of the CS solubility experiments without purification.

2.6. Characterization

The viscosities of the starch samples were measured using a rotational viscometer (Weiling Scientific Instrument Co., Ltd, shanghai, China) in a steady state at 22–23 °C. A transflective polarizing microscope (LEICA DM1000, Germany) and an SEM (F16502, PHENOM, Netherlands, 5 kV) were used to image the structure of the native CS and CS samples treated with the DES/water mixtures. The crystallinity of the starch samples was examined using an X-ray diffractometer (Ultima IV, RIGAKU) operating at 40 kV and 40 mA; the diffraction angle (2θ) ranged from 5° to 60°. To determine the structural characteristics of the alternating crystalline and amorphous lamellae, small-angle X-ray scattering (SAXS) measurements were performed with a SAXSess mc^2 (Antonpa, Austria) operating at 40 kV and 30 mA (λ = 0.154 nm). The chemical structures of the starch grains were determined using a Bruker TENSOR II FT-IR spectrophotometer. Spectra were recorded from 4000–400 cm^{-1} using 32 scans and a resolution of 4 cm^{-1}. The average molecular weight and molecular weight distribution were analyzed using Agilent PL-GPC50 and Agilent PL-GPC220 GPC. Polystyrene standards were used for calibration. All measurements were performed in DMSO containing 30 mmol/L lithium bromide as the solvent.

The CS samples treated with the DES/water mixtures were used for polarizing microscope, SAXS, and GPC analyses; and the regenerated CS was used for SEM, X-ray diffraction (XRD), and attenuated total reflection-Fourier transform infrared (ATR-FTIR) analyses.

3. Results and discussion

3.1. Solubility of starch in DESs

In a series of screening experiments, a total of 11 kinds of DESs were synthesized from different hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs) with different molar ratios of HBA/HBD. During the screening, the solubility of CS and viscosity were determined in in the different kinds of DESs (Table 1). We found that the acidic DESs typically achieved a higher solubility compared to the neutral and alkaline DESs (Table 1). Notably, solubility in the acidic DESs was not significantly affected by the molar ratios of HBA/HBD, and the highest starch solubility (52.8 wt%) was obtained with a CF molar ratio of 1:4. The lowest DES viscosity (10.85 mPa.s) was also obtained in the CF systems. The maximum starch solubility in the neutral and alkaline DESs was 11.4 wt% (EE 1:2) and 10.2 wt% (CD 1:8), respectively.

Several studies have reported that chloride anions play an important role in sugar solubility in ChCl-containing DESs (Hkkinen and Abbott, 2019; Zdanowicz and Spychaj, 2011). The N^+ of the hydrogen bond acceptor choline chloride has high electropositive activity, which can form stronger O-N.H hydrogen bonds with the -OH of starch, and gradually replaces the O-H.O hydrogen bond of starch itself, thus weakening starch hydrogen bonding. In addition, the Cl^- of choline chloride can also combine with the -OH of starch to form O-H.Cl hydrogen bonds, which further weakens the self-hydrogen bonding of starch and accelerates dissolution (Xu et al., 2012). When lactic acid,

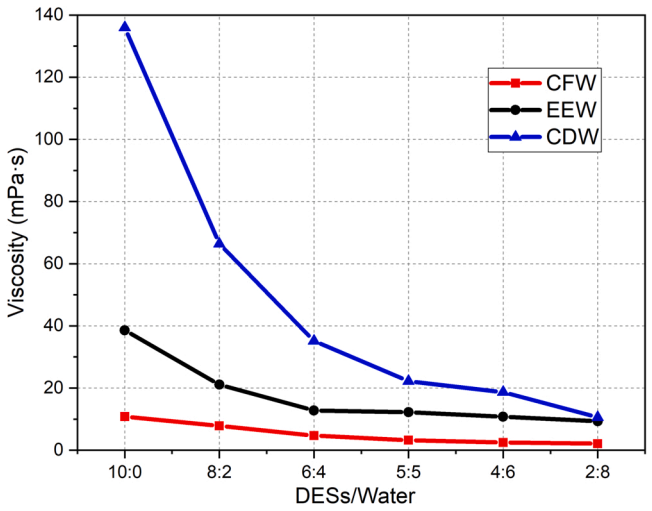


Fig. 1. Viscosity of different DESs/water mixtures (CFW, CF/water; EEW, EE/water; CDW, CD/water).

oxalic acid, and acetic acid were used as HBDs, they showed similar solubilities, whereas the solubility in CF was significantly higher. Based on previous studies, the length of the alkyl chain in ILs negatively affects the solubility of sugar (Hassan et al., 2013; Hkkinen and Abbott, 2019). Furthermore, a long alkyl chain lengths reduce the affinity with polar substances such as saccharides (Mohan et al., 2015).

Ethylene glycol DESs contain more hydroxyl functional groups of than other types of DESs, which can form stronger intermolecular hydrogen bonds, which require more kinetic energy to be broken during subsequent dissolution to achieve interaction with the solute. Diethanolamine DESs lead to excessive viscosity due to the strong interactions between their own molecular chains, resulting in low mass transfer efficiency in starch blending systems. Based on this understanding, the solubility of starch was highest in the acidic DESs followed by the neutral DESs than the basic DESs. Based on the highest starch solubility results, CF (1:4), EE (1:2), and CD (1:8) were selected from the range of prepared acidic, neutral, and alkaline DESs, respectively.

Table 1.

3.2. Viscosity of the DESs/water systems

Viscosity is one of the most influential factors affecting the solubility of starch, and hydrogen bonding is an important factor affecting the change in viscosity; therefore, the binding ability of hydrogen bonding is crucial. The electrostatic force, van der Waals forces, and hydrogen bond forces among the molecules are the main factors affecting the viscosity of the solvents. Fig. 1 shows the viscosities of the pure DESs (CFW-10:0, EEW-10:0, and CDW-10:0) were highest, with values of 10.85, 38.6, and 136 MPa·s, respectively. The viscosity of CDW decreased significantly with increasing water content, dropping substantially from 136 to 10.6 MPa·s when the CDW ratio was changed from 10:0–2:8. However, no distinct viscosity changes were observed for the EEW and CFW mixtures. The viscosity of pure EE was four-times greater than that of the 2:8 EEW mixture. The viscosities of the CFW-10:0 (pure CF) and CFW-2:8 mixtures were 10.85 and 2.1 MPa·s, respectively. At the same DESs/water ratio, viscosity was ranked in the order CDW > EEW > CFW.

DESs can not only form hydrogen bonds internally but also with H_2O. After the formation of internal hydrogen bonds in the deep eutectic system, the residual ions form hydrogen bonds with the hydroxyl groups in H_2O. Thus, there is one lone pair of nitrogen atoms and two hydrogen atoms on one amino group, meaning that three hydrogen bonds can be formed. One hydroxyl oxygen and one carbonyl oxygen can form one hydrogen bond and two hydrogen bonds, respectively. One diethanolamine molecule, which has two hydroxyl oxygen and one amino group, can form five hydrogen bonds. Similarly, one formic acid molecule has one hydroxyl group and one carbonyl group and can form three hydrogen bonds. One ethylene glycol molecule can form two hydrogen bonds. ChCl can accept three H^+ ions; therefore, three hydrogen bonds can be formed between ChCl and diethanolamine in CD, and three hydrogen bonds can be formed between ChCl and formic acid in CF. The strong interaction between the ions in the DESs leads to high viscosity, which affects the mass transfer and contact of the two phases in the subsequent reaction process (Garcia et al., 2015).

Table 2
Kamlet–Taft parameter values of the DESs/water mixtures.

	π^*	α	β
CFW-10:0 (pure CF)	1.08	1.13	0.55
CFW-2:8	1.25	1.15	0.30
EEW-10:0 (pure EE)	1.04	1.03	0.59
EEW-2:8	1.30	1.07	0.34
CDW-10:0 (pure CD)	1.01	1.00	0.76
CDW-2:8	1.34	1.02	0.31

Notes: DESs/water mixtures: CF/water (2:8) (CFW-2:8), EE/water (2:8) (EEW-2:8), and CD/water (2:8) (CDW-2:8).

The high viscosity of CDW-10:0 (pure CD) is explained by the strong interaction between the ions and the long molecular chain of diethanolamine. As depicted in Fig. 1, when H_2O is added to pure CD, the amino or hydroxyl groups of diethanolamine can form additional hydrogen bonds with water, which leads to a decrease in the viscosity of the solvent. During the extraction of polyphenols and furanocoumarins from fig (*Ficus carica* L.) leaves using DESs, the extraction rates of five target compounds, including caffeoylmalic acid, were greatly improved when water was added, probably due to the reduction in viscosity and improved mass transfer (Wang et al., 2017). The lower the viscosity of the DESs, the more efficient the mass transfer between the solute and solvent becomes. Thus, a lower viscosity can improve the permeability of the DESs to CS and accelerate dissolution; water is usually used to dilute and reduce viscosity (Ren et al., 2019a, 2019b; Zhao et al., 2015).

The Kamlet-Taft solvatochromic method was used to estimate the chemical properties of the DESs (Table 2). The Kamlet-Taft parameters α, β, and π * represent the hydrogen bond acidity, hydrogen bond alkalinity, and dipolarity/polarizability of a solvent, respectively (Xia et al., 2018; Xue et al., 2016). As showed in Table 2, the π * and α values of the pure DESs decreased and the β values increased with increasing pH, and pure CF had the highest α values, with similar reported by Oh et al. (2020), the α values of ChCl:FA (1:2) were greater than those of ChCl:EG (1:2). Acidic HBDs can effectively break the α-1,4 glycosidic bond between starches. In aqueous DESs, the acidity of hydrogen bonds did not change significantly, although the π * values increased due to the presence of water. The largest π * (1.34) was obtained in CDW-2:8. Compared with the pure DESs, β values were also decreased significantly in the aqueous DESs, which is explained by the fact that Cl^- is mostly sequestered by anionic hydrogen bonds (Ma et al., 2019).

3.3. Microscopic characterization of starch in different DESs/water systems

Starch granules typically exhibit birefringence owing to their concentric semi-crystalline layers (Shi et al., 2014). Under polarized light, “Maltese cross” patterns are formed. Therefore, observing the loss of the polarization cross under a polarizing microscope is a valuable and common method for characterizing the destruction of starch granules (Yang et al., 2019; Zhang et al., 2013).

To investigate the changes in the starch granules in the mixtures with different water contents and heating, a dissolution test was carried out using polarized light microscopy (Fig. 2). Starch (5 wt%) was dispersed in the studied DESs and heated at elevated temperatures. Different

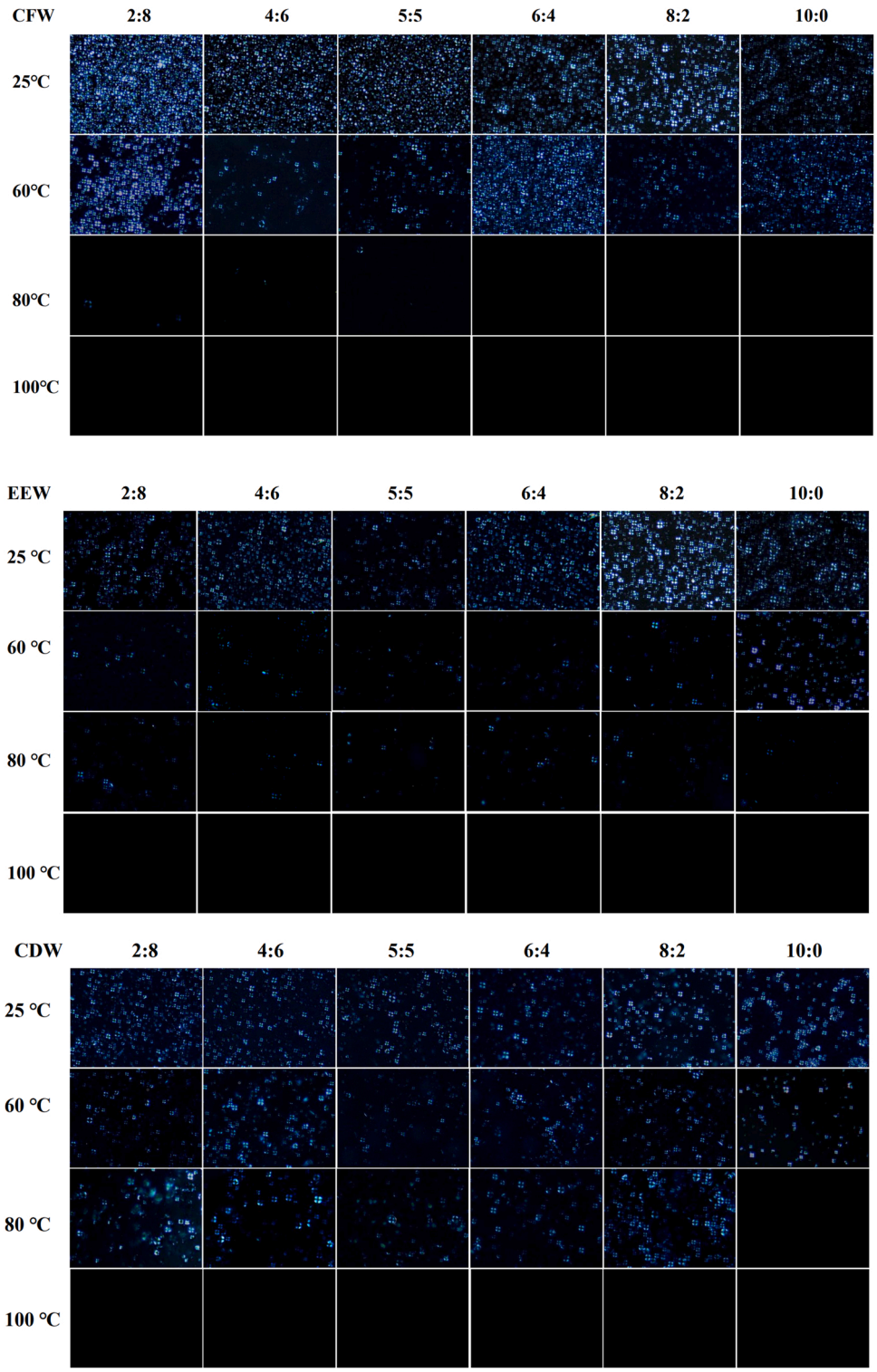


Fig. 2. Polarized light microscopy images of native cassava starch (CS) and CS samples treated with DESs/water mixtures of different mass ratios (2:8, 4:6, 5:5, 6:4, 8:2, and 10:0) at different temperatures (25, 60, 80, and 100 ℃). DESs/water mixtures: CFW, CF/water; EEW, EE/water; and CDW, CD/water.

DESs/water ratios were used, specifically 2:8, 4:6, 5:5, 6:4, 8:2, and 10:0. At a low reaction temperature (25 °C), CS treated with pure or aqueous DESs displayed a large number of clear birefringent "Maltese cross" patterns indicating that the crystalline structure of the CS granules was not amorphous at this temperature. As the treatment temperature was increased to 80 °C, the "Maltese cross" patterns disappeared and no intact granules were observed for CS treated with the CFW mixture. Similar results were observed by Mateyawa et al. (2013). At 80 °C, the granule morphology of CS treated with the EEW mixture was more disrupted compared to that of CS treated with the CDW mixture (the "Maltese cross" patterns became blurred and notably reduced). When the temperature was increased to 100 °C, all the "Maltese cross" patterns disappeared irrespective of the type of DESs used and the water content of the mixture, indicating that the efficiency of CS dissolution in aqueous DESs is highly dependent on temperature. At higher temperatures, the internal crystalline structure of starch granules is disrupted,

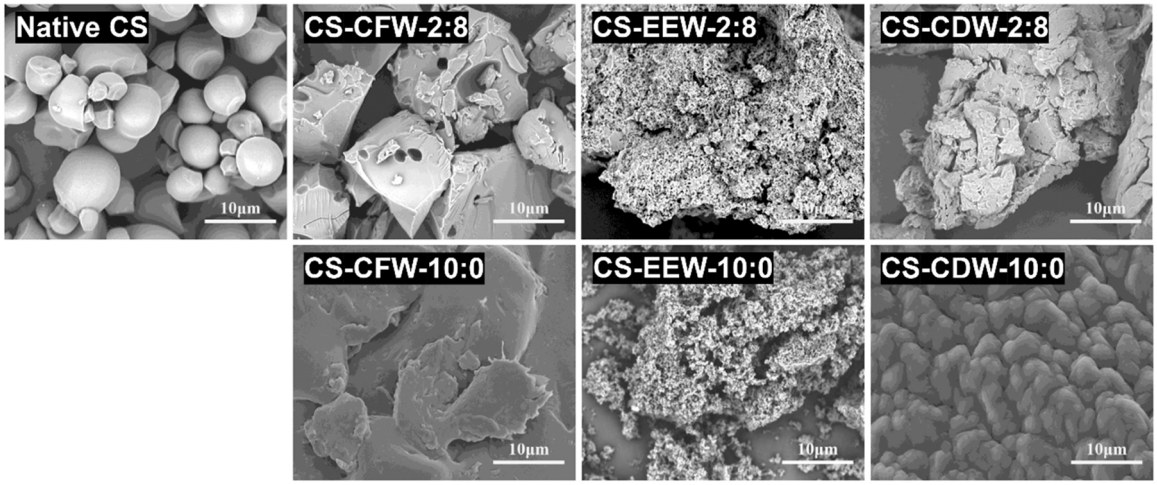


Fig. 3. Scanning electron microscopy images of native cassava starch (CS) and regenerated CS from different DESs/water mixtures (reaction temperature = 100 ℃). DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), and CD/water (2:8) (CDW-2:8), and CD/water (10:0) (CDW-10:0).

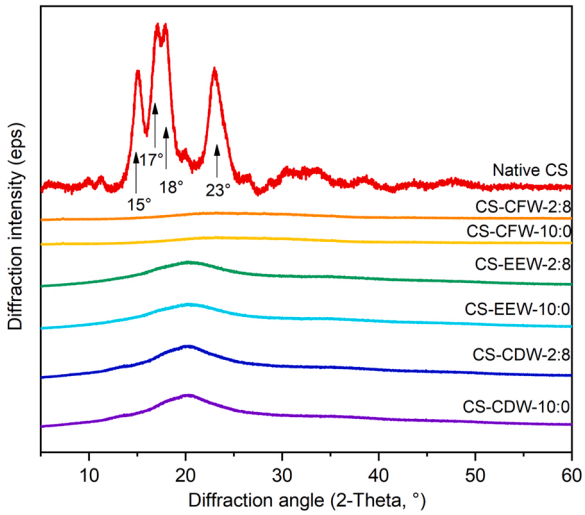


Fig. 4. X-ray diffraction (XRD) patterns of native cassava starch (CS) and regenerated CS from different DESs/water mixtures (reaction temperature = 100 ℃). DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), and CD/water (2:8) (CDW-2:8), and CD/water 10:0 (CDW-10:0).

resulting in the gradual disappearance of birefringent light (Qian et al., 2013; Wang et al., 2021).

Adding water to the DESs affected the dissolution of the starch. Compared with the pure DESs, starch efficiently dissolved during heating when the DESs/water ratios were 2:8, 4:6, or 5:5. That is, the solubility of CS in DESs could be optimized when a certain amount of water was added. Thus, aqueous DESs have lower viscosities and dissolution temperatures, biodegradability, and offer a simple means of processing of starch. These characteristics indicate that aqueous DESs offer promise for the dissolution and pretreatment of starch, which has many potential commercial applications.

The morphology and structural characteristics of the regenerated starch were analyzed using SEM. The yields of regenerated CS obtained from the different DES/water mixtures were as follows: CS-CFW-2:8 = 10.05%, CS-CFW-10:0 = 2.94%, CS-EEW-2:8 = 24.19%, CS-EEW-10:0 = 13.82%, CS-CDW-2:8 = 19.19%, and CS-CDW-10:0 = 24.79%.

Fig. 3 shows SEM images of native CS and their regenerated samples in different DESs/water mixtures. The native CS is round and has a smooth surface similar to the observations of Campanha and Franco (2011) and Sangseethong et al. (2010). After treatment with a CF/water ratio of 2:8 (sample CFW-2:8), the CS granules were destroyed and swollen. In CFW-10:0 (pure CF), all the CS granules were destroyed and aggregated, indicating that the CS was completely gelatinized. Thus, the effect of CF on the CS granules became increasingly notable at higher levels of CF.

In EEW-10:0 (pure EE), all the starch granules were destroyed, and exhibited a porous and very unsmooth morphology. The morphological change in the CS granules in the 2:8 EE/water mixture (EEW-2:8) was much less than in the EEW-10:0 mixture. After treatment with CDW-10:0 (pure CD), all the granules were destroyed and formed an unusually viscous and amorphous gel, which has also been reported in previous studies (Biswas et al., 2006; Wang and Xie, 2010).

3.4. XRD and SAXS

XRD is a useful tool for investigating the influence of DESs on the crystallinity structure of starch (Lopez-Rubio et al., 2010; Wang et al., 2019). Fig. 4 shows the XRD patterns of native CS and the regenerated samples in the pure DESs and different DES/water mixtures (2:8). The native CS shows characteristic peaks typical of A-type diffraction patterns, with distinct diffraction peaks at 15°, 17°, 18°, and 23°(2θ), which is consistent with previous reports (Kim and Huber, 2008; Monroy et al., 2018). After treatment with CFW-2:8 and CFW-10:0, these diffraction peaks became flatter, indicating a much more amorphous structure. The changes in the XRD patterns of the samples treated with the EEW-2:8, EEW-10:0, CDW-2:8, and CDW-10:0 solvents were also similar; however, the diffraction peak intensities of the CFW-treated samples were flatter than those of the other treatments. This may reflect the fact that acidic CFW is more conducive to the amorphization of starch (Wang and Copeland, 2015). Furthermore, Ren et al. (2019a) reported that the main diffraction peaks of maize starch crystallites treated with water/[C_2mim][Cl]$^-$ and water/[C_2mim][HCOO]$^-$ mixtures were gradually reduced; they became weaker and then disappeared owing to dissolution in association with [HCOO]$^-$ and [Cl]$^-$. Overall, the CS samples showed low and broad dispersion peaks for both pH changes and moisture content changes, which implies that the crystallinity of native starch was partially damaged or even destroyed during the dissolution process. The changes in the crystalline structure of the treated CS were also consistent with the described morphological changes.

SAXS technology has been widely used to characterize the semi-crystalline properties of starch granules (Blazek and Gilbert, 2011;

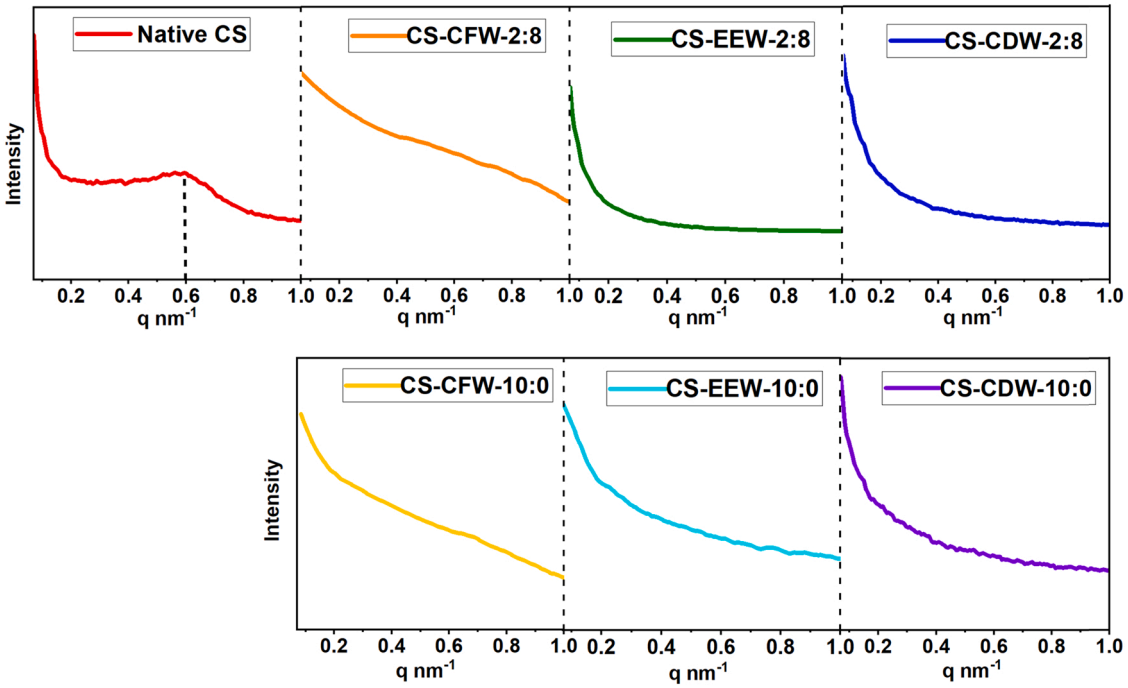


Fig. 5. Small-angle X-ray scattering (SAXS) intensity of native cassava starch (CS) and CS samples treated with different DESs/water mixtures (reaction temperature = 100 ℃). DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), CD/water (2:8) (CDW-2:8), and CD/water (10:0) (CDW-10:0).

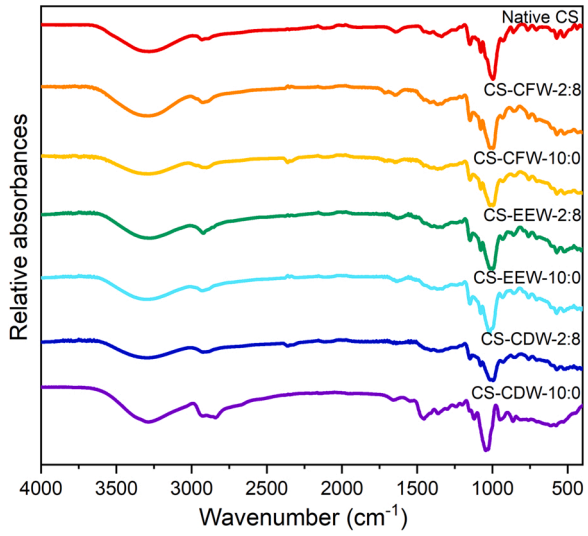


Fig. 6. Attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectra of native cassava starch (CS) and regenerated starch from different DESs/water mixtures (reaction temperature = 100 ℃). DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), CD/water (2:8) (CDW-2:8), and CD/water (10:0) (CDW-10:0).

Wang and Copeland, 2015). The SAXS results for the native and DESs-treated CS samples are shown in Fig. 5. At room temperature (22–23 ℃), the native CS samples showed a characteristic scattering peak at approximately 0.6 nm^{-1} similar to previously reported findings (Blazek and Gilbert, 2010; Cameron and Donald, 1992). For example, Bai et al. (2014) observed that when native waxy maize starch was treated with a mixture of water and glycerol (20:80 or 40:60 w/w), a characteristic SAXS peak was observed at 0.62 nm^{-1}; however, when the starch was treated with DESs, all the scattering peaks around 0.6 nm^{-1} disappeared and no other scattering peaks were observed. This indicates that when starch granules are treated with DESs, the crystalline region in the starch granules is destroyed and the semi-crystalline growth ring lamellar structure changes to a disordered state, thereby reducing the intensity of the scattering peaks and resulting in the loss of structural information (Cardoso et al., 2010). Similarly, Pikus (2005) reported that the SAXS curve of treated native potato starch was notably smooth, and the starch molecules were disorderly arranged.

Table 3
Molecular weight distribution of native cassava starch (CS) and CS samples treated with DESs/water mixtures (reaction temperature = 100 ℃).

	M_n (g/mol)	M_w (g/mol)	M_w/M_n
Native CS	9147	89,179	9.75
CS-CFW-2:8	988	2290	2.32
CS-CFW-10:0	138	231	1.67
CS-EEW-2:8	860	1896	3.37
CS-EEW-10:0	231	745	3.23
CS-CDW-2:8	1448	2811	1.94
CS-CDW-10:0	405	1066	2.63

Notes: M_n, number-average molecular weight; M_w, weight-average molecular weight; DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), CD/water (2:8) (CDW-2:8), and CD/water (10:0) (CDW-10:0).

3.5. ATR-FTIR

ATR-FTIR is commonly used to characterize the chemical structure of starch and is an effective tool for investigating the interactions between DESs and starch granules (Qi et al., 2019; Sevenou et al., 2002). The ATR-FTIR spectra of the native CS and regenerated starch are shown in Fig. 6, and the signal assignments are listed in Table S1. The spectra of all the starch samples showed no significant changes after DES treatment, indicating that there was no reaction or derivatization of starch with the DES components. Compared to the native CS samples, the intensity of regenerated starch notably decreased owing to its amorphous

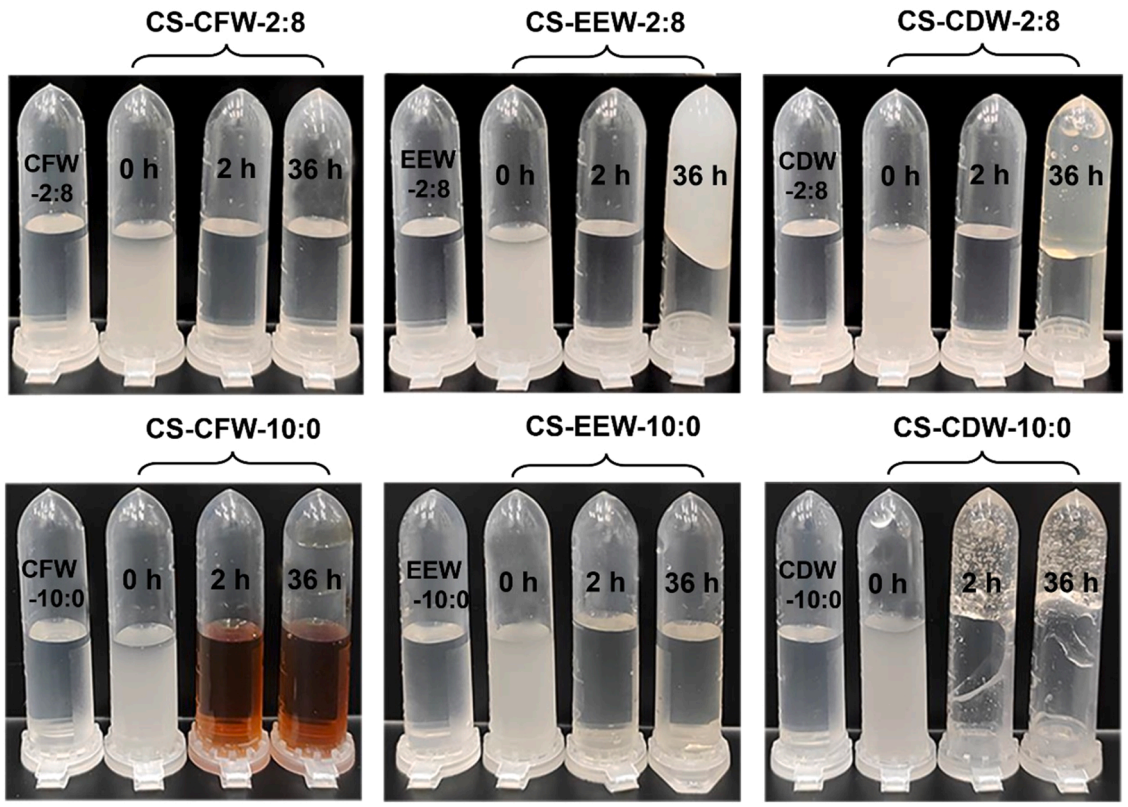


Fig. 7. Retrogradation photographs of cassava starch (CS) samples treated with DESs/water mixtures (DESs/water mixtures: CF/water (2:8) (CFW-2:8), CF/water (10:0) (CFW-10:0), EE/water (2:8) (EEW-2:8), EE/water (10:0) (EEW-10:0), CD/water (2:8) (CDW-2:8), and CD/water (10:0) (CDW-10:0); at 0 h, 5 wt% CS was added into the DESs without heating; at 2 h, the CS-DESs samples were heated for 2 h at 100 °C; at 36 h, the CS-DESs samples were heated for 2 h at 100 °C and stored for 36 h at 22–23 ℃).

structure. Similar results were also reported by Zdanowicz et al. (2016).

3.6. GPC

The molecular weight distributions of the native and DESs-treated samples were determined by GPC (Table 3). The weight-average molecular weight (M_w) of the native CS was 89,179 g/mol, and the M_w to number-average molecular weight (M_n) ratio was 9.75. After treatment with different DESs/water mixtures, the molecular weight distribution decreased sharply, indicating the cleavage of glycosidic bonds. The M_w of the pure DESs (CFW-10:0, EEW-10:0, and CDW-10:0) was 231, 745, and 1066 g/mol, respectively. The molecular weight of the CDW-treated samples was higher than that of the CF and EE DESs-treated samples, which indicates that the glycosidic bonds were more stable under alkaline conditions. This is in agreement with the previously described results of the dissolution tests. M_w increased significantly with increasing water content in the DES mixtures; the M_w of CFW, EEW, and CDW was 2290, 1896, and 2811 g/mol, respectively, when the DESs/water ratio was changed from 10:0–2:8. When DESs was added with the corresponding proportion of water, the molecular weight of CS dissolved by the aqueous DESs was larger than that dissolved by the pure DESs. The presence of a moderate amount of water can reduce the by-products of the further polymerization of glucose, dehydration, etc., thus inhibiting the formation of humins and improving the glucose yield (Binder and Raines, 2010). Furthermore, this plays a regulatory role in starch modification processing by controlling moisture content.

The M_w/M_n ratio is the polydispersity coefficient, which reflects the narrow distribution of the molecules. The M_n and M_w distribution of CS can be used to characterize its solubilization by different DESs/water mixtures (Torres et al., 2020). Here, the M_w/M_n ratio of the native starch was 9.75; the lowest value (1.67) was obtained for the CFW-10:0 (pure CF) treatment sample. This indicates that the pure CF-treated sample had the lowest polydispersion and the most uniform molecular mass distribution. The amylose content of CS ranges from 15.2% to 26.5% (Fan, 2015), which indicates that amylopectin is the main component. Compared with high-amylose starch, starch with a low amylose content is more susceptible to acid hydrolysis. Therefore, CS is considered highly susceptible to acid hydrolysis, during which the glycosidic bonds of amylose and amylopectin molecules break (Gérard et al., 2002), resulting in a rapid decline in M_w and M_n.

3.7. Starch retrogradation

Starch retrogradation refers to the transformation from a disordered state to an ordered crystalline state, which for starch is often affected by the molecular structure, temperature, and other conditions (He et al., 2014). However, retrogradation often leads to changes in the hardness, viscosity, and mechanical properties of starch, as has been systematically analyzed in numerous studies (Han and Lim, 2004; Jayakody and Hoover, 2002; Karim et al., 2000). Therefore, the regeneration of CS after treatment with different DESs was also explored as part of this study.

Fig. 7 shows that the six prepared DESs were clear liquids, as has been reported elsewhere (Farooq et al., 2020); however, the mixtures became turbid after CS was added. As the temperature was increased to 100 °C, discoloration was observed for the CFW-10:0 (pure CF) mixture, whereas the CFW-2:8 mixture remained a clear, colorless liquid (Fig. 7). This indicates that the acidity of pure CF may lead to side reactions. Similar browning effects have been reported for IL-dissolved carbohydrates (Hkkinen and Abbott, 2019).

In the EEW-10:0 (pure EE) mixture, CS formed a clear, transparent, and stable liquid at 100 °C; changes in transparency and color were not

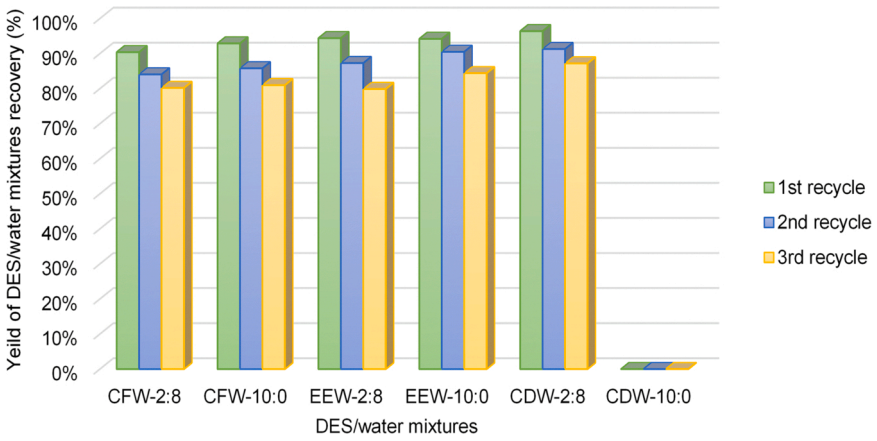


Fig. 8. Recycling of DESs/water mixtures used for the dissolution of native cassava starch (CS).

observed. In contrast, while starch formed a transparent gel in the EEW-2:8 mixture at 100 °C, a white-colored flocculate formed after 36 h of storage. This indicates the regeneration of starch in the EEW-2:8 treated samples. Moreover, the CDW-10:0 (pure CD) mixture formed a highly viscous liquid following the starch treatment, which is consistent with the viscosity test results for the DESs (Table 1). However, when the starch was treated with the CDW-2:8 mixture for 2 h at 100 °C, a clear liquid was initially formed, which turned into a cloudy gel after 36 h. Thus, the rate and extent of starch retrogradation were largely dependent on the mixture water content (Wang and Copeland, 2013; Wang et al., 2015).

3.8. DESs/water mixture recycling and reuse

To make the DESs/water mixing process more practical for industrial application, recovery and recycling capacity must be considered. Thus, recyclability of the DESs was investigated based on the reuse of the mixtures. As shown in Fig. 8, the recovered DESs maintained good CS solubility (approximately 80%) after being reused three times. However, CDW-10:0 showed poor reusability due to its excessive viscosity.

4. Conclusions

The solubility of CS was explored in a series of acidic, neutral, and alkaline DESs. Three typical DESs (CF, EE, and CD) were selected and used to determine the effects of water content on their ability to dissolve CS. Based on our results, a high water content corresponded to low viscosity, and a high temperature corresponded to strong dissolvability. All three types of DESs were found to dissolve CS to varying degrees, and optimal solubility was obtained when a certain amount of water was included in the mixtures. Among the three tested DESs, the ChCl/formic acid mixture achieved the best dissolution effect, and the starch was not retrograded. Based on our findings, the composition of DESs and the water content should be considered in the process of dissolving starch, as these are crucial factors affecting subsequent mass transfer and regeneration. Overall, these findings provide valuable information to support the development of sustainable and inexpensive solvents, expanding the potential industrial application of DESs for starch, such as the conversion of reducing sugars and hydroxymethylfurfural, as well as the development of starch-based conductive materials.

CRediT authorship contribution statement

Lingyu Zeng: Conceptualization, Data curation, Formal analysis, Writing – original draft. **Wei Mao**: Data curation. **Jiawen Hao**: Investigation, Reviewing. **Guirong Ye**: Investigation. **Xueping Song**: Software. **Ling Zeng**: Reviewing. **Shuangfei Wang**: Supervision, Methodology, Reviewing. **Jinghong Zhou:** Conceptualization, Methodology, Reviewing and Editing, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2021.114453.

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